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RIVERS OF CONVERSATIONS
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What is Responsible Design Education?

Luke Feast, Bo T. Christensen, Peter Gall Krogh, Colin M. Grey, and Linda Nhu Laursen

From being an ideal, it has become imperative to act responsibly in all matters in society. While responsible design is described as responding to the needs of and challenges faced by society, the understanding of what a responsible design education entails is still unclear.

In this paper, we investigate the scope and meaning of "responsible design" in design education through a mixed-methods study incorporating a scoping review and a participatory workshop. The scoping review analysed 62 studies on responsible design education covering dimensions including ontology, outcomes, audience, assessment, content, and pedagogy. The participatory workshop involved 12 design academics sorting and discussing key quotes extracted from the literature.

The findings suggest responsible design education should develop students' ethical awareness and an understanding of design's systemic impacts. However, there is tension between prioritizing user needs versus broader systemic effects and between ethical reflection and technical production. The content may include ecological literacy, stakeholder values, ethical frameworks, and professional ethics. Pedagogy emphasises real-world projects, stakeholder collaboration, workshop activities exploring ethical dilemmas, and scenario-based activities.

We suggest that future research should investigate institutional factors shaping social accountability and develop tailored assessment frameworks. Overall, the paper advances the understanding of responsible design education to equip future designers to address complex socio-ecological design challenges.

KEYWORDS: responsible design, design education, responsibility, ethics, literature review, participatory workshop, mixed-methods research

RSD TOPIC(S): Learning & Education, Socioecological Design

Introduction

Today, we live in a world facing many urgent design challenges related to issues such as sustainability, gender, rights, inclusivity, marginalisation, peace, corporate social responsibility, climate, food, and energy. Thus, from being an ideal, it has become imperative to design responsibly in all matters in society. However, this societal responsibility is not new, it can be traced back to the origin and evolution of contemporary design research, which has been shaped by cognitive science, that emphasises the limitations of bounded rationality and cognitive resources of the design process (Pask et al., 1962; Simon, 1969).

Historically, when the socially responsible design movement emerged in the late 1960s and early 1970s (Cross, 1972), Schön's concept of design as a reflective practice (1984) introduced ethical considerations into the design process. Also, important figures such as Victor Papanek (2000) recommended designers and stakeholders should address real social needs by collaborating through dialogic and participatory design processes.

In the past decade, an ontological shift has occurred, democratising design and framing it as an innate human capability rather than an exclusive professional domain (Manzini, 2015). Design research disciplines, topics and methodologies have broadened and diversified (Tollestrup et al. 2023; Krogh & Koskinen, 2020; Christensen & Ball, 2019). A

paradigm shift that necessitates an ethical re-evaluation of design practice, research, and its cultivation as a universal human responsibility (Ozkaramanli et al. 2024).

The rise of sustainability and inclusivity as key design agendas has further redefined the role of the user, expanding it to encompass broader concepts of empathy and interconnected value chains (Moreno et al., 2016). This shift emphasises the development of responsible design practices that extend beyond human preferences to promote overall global well-being (Escobar, 2018; Wakkary, 2021). While the term 'responsible design' is described as design that "responds to the needs of and challenges faced by the society" (Eggink et al. 2020), the understanding of what a responsible design education entails is still unclear.

Currently, in the sense of urgency, there is an overwhelming set of initiatives, challenges and re-orientations of previous approaches being both arbitrarily and systematically discussed within design ethics literature, including but not limited to sustainability, gender issues, ethnic rights, inclusivity, economic marginalisation, reconnecting man and nature, peace movements, CSR and business ethics, climate crisis responses and food and energy scarcity (Vesti et al., 2024). Despite these overwhelmingly diverse discussions, we lack a clear conceptual understanding of what 'responsible design' is, its implications and how it may be practised.

We aim to develop an understanding of what responsible design education is, how it may be taught, and what its implications are for educational institutions and the design discipline. Through dialogue and conversation, ethics and responsibility resonate within systems thinking and design thinking. Heinz von Foerster (2003) wrote that objectivity is a "device for avoiding responsibility" (p. 293), whereas ethics "manifests itself through dialogue" (p. 297). In this paper, we seek to further such ethical dialogue, which scholars have called for in design education (Vesti et al. 2024).

We present a mixed-methods study incorporating a scoping review and a participatory workshop. Our guiding research question is: What is the scope and meaning of “responsible design” in design education? We will answer this research question by:

- Charting the bibliographic characteristics of the 62 studies included in the scoping review (Table 1)
- Reporting findings from the literature and participatory workshop using Fayolle and Gailly’s (2008) teaching model framework
- Discussing key findings and a potential gap in the literature regarding assessment procedures for responsible design education, as well as connections between various factors, such as design sub-domains, degree types, design paradigms, institutional contexts, and geographic regions, in shaping social accountability in design education and professional practice.
- Suggesting future research to investigate how various institutional factors shape social accountability in design education.

Research design

We employed a mixed-methods approach, integrating data from a participatory workshop with expert design educators and a literature review of responsible design education in published research and scholarship. We used this research design to situate our primary data collection in the literature while incorporating collaborative insights from design education experts. We integrated insights from both methods as a means of methodological and data triangulation, enhancing the robustness and trustworthiness of our analysis and conclusions.

Although we did not register a formal literature review protocol, we followed the PRISMA-ScR (Tricco et al., 2018) scoping review parameters and methods throughout the review process. We included peer-reviewed journal articles and targeted peer-reviewed conference papers written in English, published between 1999 and 2024. Studies had to relate to design education (rather than design practice) and primarily focus on teaching, outcomes, and conceptualisation of "responsible design education." We excluded studies solely focusing on education in non-design fields, such as fine arts, management, psychology, and engineering, to ensure a manageable task within the available time frame.

We searched the Web of Science database, the first two pages of results in Google Scholar, and targeted design journals and conferences (see Appendix 1 for the full list).

Our search strategy in Web of Science included the following key parameters: title containing "Responsib*" or "ethic*", topic "Design education", publication date from 1999 to 2024, English language, and document types limited to articles or proceedings papers. The most recent search was executed on 7 March 2024.

We screened the records by reviewing titles, keywords, and, if necessary, abstracts to determine eligible studies (Figure 1). From an initial 110 records, we excluded 12 duplicates, three records based on type, three due to inability to retrieve full texts, and a further 26 reports that did not primarily relate to teaching, outcomes, or conceptualization of responsible design education. This process yielded 62 studies for inclusion in the review.

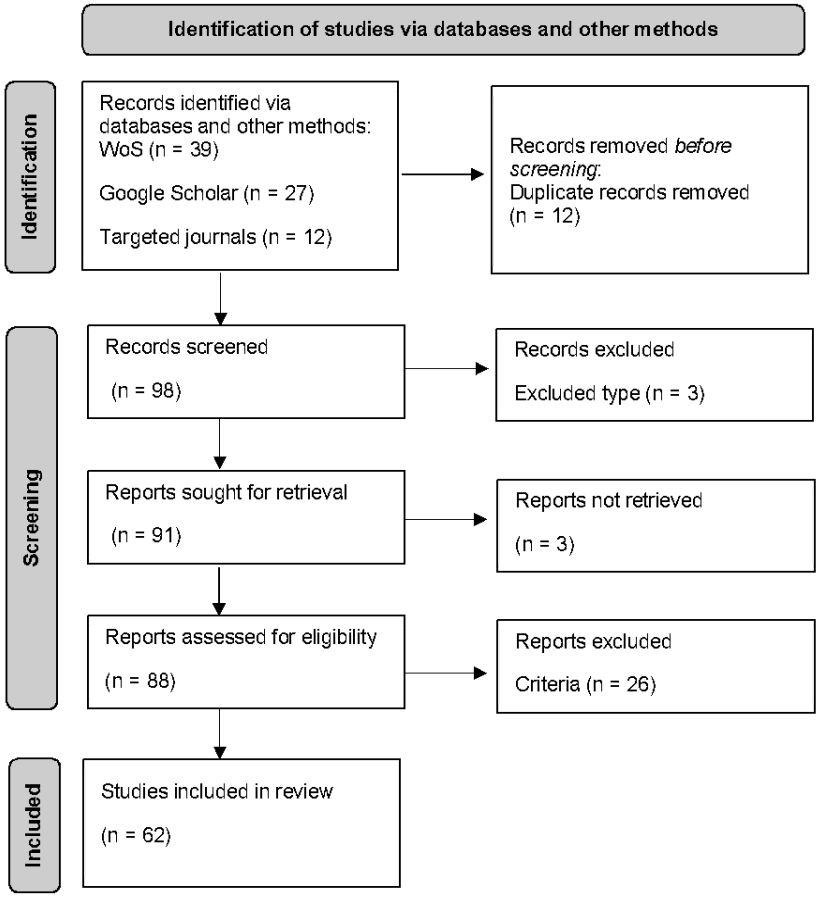


Figure 1 The process and results of literature identification and selection

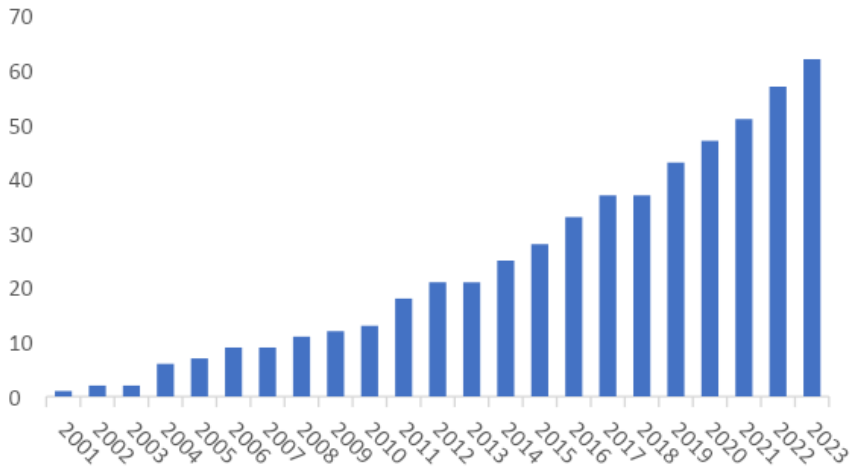


Figure 2. Cumulative frequency of studies by year published

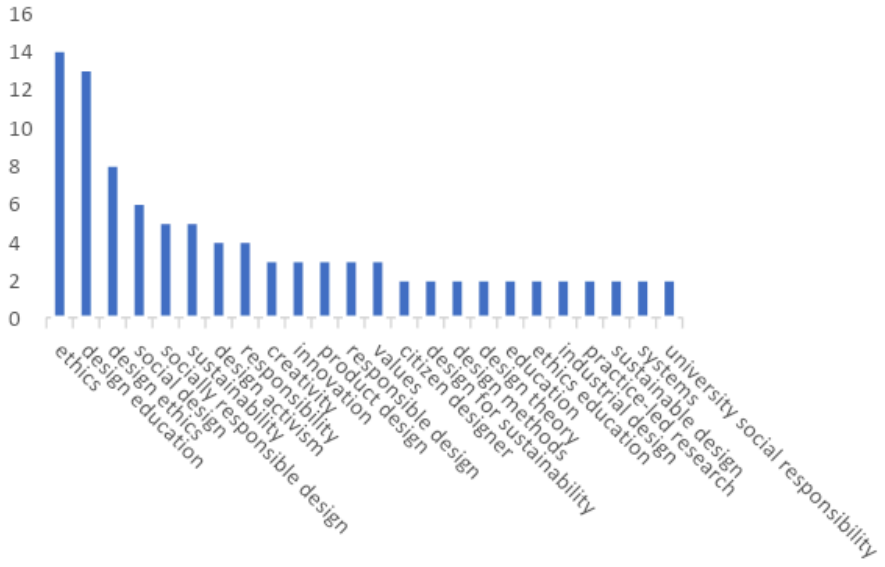


Figure 3. Frequency of author-supplied keywords (n ≥ 2)

A researcher charted the data using an Endnote database. They extracted each study's bibliographic information, field of research or practice, author affiliation country, author-supplied keywords, and key findings related to responsible design education.

The cumulative frequency shown in Figure 2 indicates a steady rate of publication. Figure 3 shows the most frequent author-supplied keywords related to design education and ethics, social responsibility, sustainability, and activism.

Table 1 shows the characteristics of the studies included in the review. The frequency of studies published per year increased from n=6 between 2000-2004 to n=19 between 2020-2024. 50% of studies included in the review had first authors affiliated with institutions in the US, Australia, or the UK. 52% of the studies' fields of research or practice were either interdisciplinary or unspecified rather than focusing on a specific field of design such as industrial design, HCI, or architecture.

We synthesized the data using Fayolle and Gailly's (2008, p. 572) teaching model framework, which encompasses an ontological level and five interrelated educational dimensions: Why?, For whom?, For which results?, What?, and How? Our synthesis followed a four-stage process: compiling literature in NVivo software, disassembling through reading and coding, reassembling by grouping similar concepts within the framework and interpreting by integrating insights from both the scoping review and the participatory workshop.

We decided to use Fayolle and Gailly's (2008) teaching model framework because of its comprehensive and multidimensional nature. The framework's ontological and educational dimensions provide a holistic perspective that captures detailed aspects, allowing for a systematic synthesis of the findings from the workshop and literature. However, we recognise that the framework's focus on formal elements may not fully account for informal experiential learning in design education. Furthermore, the framework's focus on educational dimensions might underemphasize the role of other factors such as design paradigms, institutional contexts, and geographic regions, in shaping social accountability in responsible design education. Nevertheless, despite these limitations, Fayolle and Gailly's (2008) teaching model provides a useful framework for interpreting the findings while leaving room for future research to investigate more nuanced aspects of responsible design education.

CATEGORY	CHARACTERISTIC	FREQUENCY	%
Year of publication	2020-2024	19	31%
	2015-2019	18	29%
	2010-2014	13	21%
	2005-2009	6	10%
	2000-2004	6	10%
Publication Type	Journal Article	32	52%
	Conference Paper	30	48%
Country of First Author	USA	16	26%
	Australia	8	13%
	UK	7	11%
	Netherlands	5	8%
	Norway	4	6%
	Taiwan	3	5%
	Iran	2	3%
	Canada	2	3%
	Denmark	2	3%
	Brazil	2	3%
	Turkey	2	3%
	Japan	1	2%
	Singapore	1	2%
	Mexico	1	2%
	Germany	1	2%
	Sweden	1	2%
	United Arab Emirates	1	2%
	Ireland	1	2%
	Croatia	1	2%
	Belgium	1	2%
Field of Research or Practice	Interdisciplinary or Unspecified	32	52%
	Industrial Design or Engineering	11	18%
	HCI, IxD, UX, or ICT	6	10%
	Architecture, Interior, Urban	5	8%
	Human-Centred Design	3	5%
	Visual, Graphic, or Media Design	3	5%
	Service System Design	1	2%
	Textile and Fashion	1	2%

Table 1. Study Characteristics

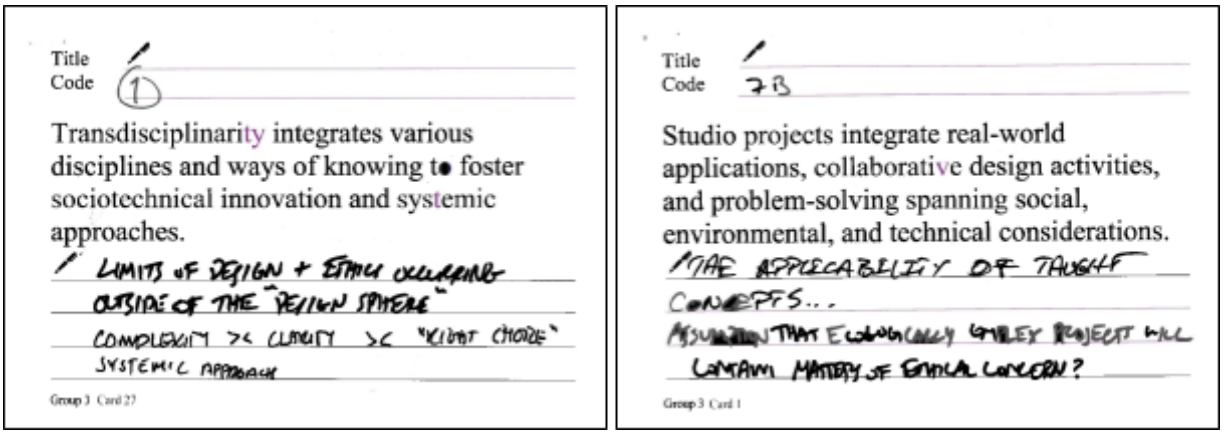


Figure 4. Examples of workshop cards

Findings

Ontology: Defining responsible design education

The participatory workshop revealed diverse perspectives on the distinction between design ethics and applied ethics. One participant highlighted the difference between ethical considerations within design practice compared to broader applied ethical principles. Pointing out it was interesting that Chan (2006, p. 3548) defined “the ethics of design as different from applied ethics… in this case, they were saying there's something unique about the generative nature of design and design ethics that we need to understand" (Transcript 3, 23:40 to 23:24). Another participant asked, "I guess the big question is what's the difference between design and responsible design? And is there a difference? Shouldn't all design be responsible?" (Transcript 2, 14:39 to 14:51). They challenged the conventional understanding of design and suggest responsibility is embedded within the design process and the agency of design.

Furthermore, yet another participant reflected on students' attitudes towards responsibility in design. They said, "We had a discussion with some of my students, they admittedly say they do irresponsible design because it's just for commercial purposes, I just want to earn money. They have awareness but they don’t care. So, is that responsible or irresponsible, or where are we positioned?" (Transcript 1, 01:09:11 to 01:09:47). This anecdote accentuates the ethical dilemmas faced by designers and educators, particularly in balancing commercial interests with ethical considerations.

The literature emphasises ethical principles in design education. Ozkaramanli et al. (2022, pp. 9-10) locate responsible design in the notion of taking care in every decision in collaborative and relational aspects of ethical inquiry. While Elliott (2004, p. 124) claims an ethical stance requires empathy towards all living beings, including non-human living systems, future generations, and the planet itself. Implying the responsibility of designers to be the voice of the voiceless.

Bousbaci and Findeli (2005, p. 253) frame responsible design as praxis, where designers see their role as an ethical, reflective practice rather than just technical production (also, see Findeli, 2001, p. 14). According to Mohamed (2021, p. 2), embedding ethics in design education should provide rules for promoting ethical decision-making. Developing moral consciousness through attending to matters of concern, Stewart and Lorber-Kasunic (2006, p. 238) argue, enables the designer to intervene to do what users know they ought to do.

Design ethics is the “study of morals and morality in design practices," according to Chan (2016, p. 3539), "which encapsulates knowledge that evaluates, justifies and guides design.” Fry (2004, p. 152) argues that ethical design requires thinking, modelling, and projecting consequences so as “to know what to create (as means to sustain futures wherein the responsibility for being anthropocentric is accepted as the basis of being another way) and knowing what to destroy.” Grey and Boling (2016, p. 974) see ethics and values combined within professional disciplinary identity and personal identity, actions, and behaviour. While Davis and Dubberly (2023, pp. 110-111) argue designers must take a broad, systemic perspective considering all interdependencies and potential consequences of their work across social, environmental, and technological systems.

“Socially responsible design (SRD),” write Cipolla and Bartholo (2014, p. 90), “means adopting design practice—or more precisely, transformation design—to promote change in local contexts through the development of regenerative solutions.” They claim socially responsible design necessitates participatory methods, multidisciplinary collaboration, and co-creating solutions with users/stakeholders. According to Melles et al. (2011, p. 149), socially responsible design should empower communities, develop local capabilities, and focus on addressing the real needs of users/communities,

especially underserved populations, rather than just creating products for profit motives.

Interestingly enough, as voiced by the participants in the workshop and as pointed to in Scuri et al. (2022) and Besana et al. (2024) articulating and supporting economic and commercial interests in design and design research is marginalized and often disregarded regarding sustainability and responsibility rather than being a vehicle for the objectives. With the extension of socially concerned and participatory practices gaining momentum in design over the past four decades, there is a shift in the discourse defining responsible design education towards emphasising more-than-human, ethical, reflective practice complementing classical technical production, while also marginalizing that design solutions should be economically viable. However, curiously enough, all practising designers are aware that there is no such thing as viable design without considering how a product, service or system can be sustained also economically.

Why?

The "Why?" dimension includes various perspectives on the purpose and outcomes of integrating ethics into design practice.

Workshop participants questioned traditional design education's focus on technical skills without considering the broader societal implications. One group questioned whether responsibility "is in the product or it's in the person or it's in the industry... would you have any part of your education that would promote that kind of design that would not have a responsible element?" (Transcript 2, 48:31 to 49:05). This view suggests a shift towards purposively including ethics and responsibility in design education.

Another participant questioned ethics' origin, stating, "The difference between ethics as something that is originating from theory, philosophy, et cetera. Or is ethics and responsibility something that is done or something that's carried out?" (Transcript 1, 01:03:26 to 01:04:16). This observation highlights the distinction between theoretical ethics and the practical application of ethical principles in design practice.

Participants also questioned the purpose of engaging with ethics within design education. One participant pondered about the teleology or praxis of ethics, "So what is

the purpose of engaging in this ethical conversation? Is it a social good? Is it living the good life?" (Transcript 3, 18:27 to 18:40). This emphasises clarifying the goals and motivations behind integrating ethics into design education.

The literature provides insights into the outcomes of responsible design education. According to Das et al. (2023, p. 12), the outcome is that designers consider ethics, equity, and justice in their design work, for example, the inclusion of historically marginalized groups in engineering. While Grey et al. (2023, p. 2387) suggest that improving the practice resonance of ethical design methods may support ethical awareness and action.

Ethical awareness involves understanding ethical responsibilities, which include, according to Haug (2017, pp. 658-660), considering environmental, social, and economic impacts throughout the lifecycle of a design object. Khaniki and Khalili (2017, p. 4) explain ethical awareness requires considering impacts on humans, animals, and the environment across the production, use, and disposal phases. Furthermore, as Sonneveld (2014, p. 1) relates, ethical awareness is related to critical reflection on one's moral framework or, according to Grey et al. (2023, p. 2380), one's ethical commitments, such as the Hippocratic Oath.

Perspective-taking, according to Debs et al. (2023, pp. 159-160) entails understanding and integrating diverse viewpoints into the design process. For Af Ornäs and Keitsch (2016, p. 3), this involves pointing out alternative interpretations and societal impacts, encouraging reflection on personal ethics and professional ethics, or, as Grey et al. (2023, p. 2379) suggest, understanding user and stakeholder needs and experiences.

Grey (2019, p. 309) argues that integrating ethics into the design process can develop an expanded understanding of the problem space to infrastructure or system-level issues. Furthermore, Grey et al. (2023, pp. 2379-2380) found abilities including assessing goals and evaluating outcomes through various ethical lenses, supporting reflection and consideration of the implications. Additionally, Khaniki and Khalili (2017, p. 3) describe behaviour change models that may be used to guide users towards responsible behaviours.

Problem framing and recognising dilemmas, according to Af Ornäs and Keitsch (2016, p. 6), addresses design students' ethical consciousness, personal ethics, and empathy. As

Grey (2019, p. 309) describes awareness of the ethical role in framing a problem strengthens designers' argumentation skills in discussing ethical paradoxes and impacts (also, see Grey et al., 2023, p. 2380). Furthermore, according to Grey et al. (2023, pp. 2379-2380), value identification assists designers to incorporate values into their design processes and products.

The "Why?" dimension suggests a need to clarify the overarching goals and motivations behind integrating ethics into design education.

For Whom?

The "For Whom?" dimension of responsible design education explores the various stakeholders to whom designers are accountable and the ethical considerations that guide their actions. Whereas the framework by Fayolle & Gailly (2008) mainly explores variability in student populations ('who are the students'), we found it pertinent and meaningful to analyse 'whom' design education is considered socially responsible towards.

The reviewed literature as well as workshop participants pointed towards a diversified set of targets of social responsibility in design.

Workshop participants raised questions about the recipients of designers' responsibilities, stressing the tension between prioritizing user needs versus considering broader systemic impacts. One participant remarked, "There is also a question of who are you responsible to or for. So, some cards emphasise user needs... while other cards mention something like, I don't know, nonhumans or ecosystems that are more systemic components" (Transcript 2, 03:24 to 03:55). This observation identifies designers' needs to navigate complex ethical considerations that extend beyond immediate user concerns and illustrates that the targets of social responsibility in design education include both immediate stakeholders with whom the designer directly engages, as well as indirect and broader social systems and global issues.

Another participant pondered the relationship between personal and professional ethics, questioning which ethical frameworks should guide design education. They remarked, "What is the relationship between personal and professional ethics? And which are we educating for?" (Transcript 3, 50:43 to 50:56). This highlights the

importance of distinguishing both personal values and professional ethics in design education to develop a broad understanding of responsible design.

The literature offers insights into design students' ethical awareness and responsibility. According to d'Anjou (2004, p. 217), awareness and self-creation are when students become conscious of their engagement towards people and the world. They can grasp the complex and dynamic situation of a problem and the consequences on the natural, social, cultural, and artificial worlds. Furthermore, d'Anjou (2004) explains the aim:

> "Students develop a "habitus" that leads to an attitude when engaging in a professional problem. In the larger scope the goal of professional education is to have students develop a general attitude and a global judgment at the moment of engaging problems." (p. 216)

Virtuous character in design, according to Haug (2017, pp. 657-658), draws on virtue ethics and the development of moral character, particularly highlighted by Aristotle's ethos of phronesis (knowledge and skills), arete (morally good intentions and actions), and eunoia (goodwill and ability to engage others). For Grey (2019, p. 301) the ethically aware design character is a subset of 'design character' involving the framing process, navigating complex ethical dilemmas, acknowledging diverse stakeholder needs, and continuously reflecting on one's design philosophy (also, see Grey & Boling, 2016, pp. 978-979).

Establishing deliberation phases in design processes, according to d'Anjou (2004, p. 215), supports designers to integrate reflective and ethically responsible decision-making in design. Watkins et al. (2020, p. 2115) discuss enacting one's design philosophy in both negotiating competence and the capacity to negotiate one's competence with others. They highlight the tensions that arise between individual design philosophies and organizational contexts, suggesting that understanding these tensions can inform career decisions and opportunities for advocating user-centred approaches in design.

Several authors state that praxis is where ethical considerations and responsibility play crucial roles. Bousbaci and Findeli (2005, p. 257) advocate for acknowledging all actors as "projecting beings," capable of initiating actions aimed at enhancing the well-being of others. Similarly, d'Anjou (2004, p. 213) stresses that professional design practice both

shapes the external world and personal identity. Lloyd and van de Poel (2008, pp. 445-446) highlight practical experience in ethical decision-making, emphasising the need for designers to "feel" ethical concepts alongside theoretical understanding. According to Sonneveld (2014, p. 6), positive ethics provides students with a supportive and inspirational vision of ethics in designing, rather than intimidating or restricting them.

Taken together, the literature review and participatory workshop point towards a need to consider a variety of stakeholders with whom the designer holds both direct and indirect relationships. The findings suggest that the 'whom' the designer is socially responsible towards can be distributed along a dimension of immediateness or 'directness of interaction', extending from immediate interactions (e.g., users, clients), to both organisational and cultural others with whom the designer may not immediately interact, and finally to global issues (e.g., sustainability). Further, the literature suggests the integration of personal values and professional ethics.

What?

The "what" dimension explores the essential components and perspectives of responsible design education.

One participant emphasised, "Design practice needs ecological literacy... but it's also inviting... the development of the individual... building personhood... being a reflective being" (Transcript 3, 38:50 to 39:49). This perspective encourages integrating environmental awareness and personal ethical development into responsible design education.

The literature echoes the participants' questioning, with scholars advocating for interdisciplinary collaboration and ecological literacy in addressing sustainability challenges. For example, environmental sustainability, as per Boehm's (2015, p. 77), TIER (Teaching Interdisciplinary Environmental Responsibility) model addresses sustainability by incorporating multiple academic disciplines. Boehnert et al. (2022, pp. 6-7) emphasise ecological literacy in design practice, advocating for the interdependence between natural processes and human activities that enable sustainable transitions. Many sources highlight the socioeconomic and environmental impact of sustainable development to improve quality of life and reduce energy consumption, for example in

urban design (Khatami et al., 2023, p. 267), textile and fashion design (Leerberg et al., 2010, p. 309), and product design engineering (de Vere et al., 2009, pp. 3-4).

Another participant questioned the significance of empathy in responsible design, stating, "If you are responsible, you need to have not only broad knowledge, you need to have empathy... [But] you could also say that the role of a designer is just to generate profit" (Transcript 3, 18:00 to 18:33). This points to the questioning of the assumption that empathy is a foundational principle in responsible design practice, as most professional design runs under the regime of capitalist thinking. Educating designers and executing design often puts the designer in a quandary balancing economic benefits and user-perceived usefulness.

Design education requires ethics and values, Donahue (2004) argues, to shift the focus from technical considerations to broader ethical principles related to praxis. Culén and Karahasanovic (2022) emphasise understanding stakeholder values and Ferrarello (2019) wants responsible design education to consider cultural and historical contexts. Meanwhile, Godbold et al. (2019) aim for design students to consider human-centred design projects' ethical implications and prepare formal ethics applications for research. Sonneveld (2016, pp. 2-4) focuses on developing ethical sensitivity to recognise ethical aspects in design projects, analysing them systematically, and making informed moral judgments.

The literature introduces various ethical perspectives, including virtue ethics, deontology, utilitarianism, care ethics, and common-sense ethics. Scholars explore ethical perspectives in design from different levels: universal values, professional ethics, and personal values. This includes considering universal human rights, developing professional ethical codes, and aligning personal values with professional practice. Yürekli (2015, pp. 79-80) extends the discussion to philosophical considerations such as truth, meaning, politics, power, the relationship between values and design decisions, and practical dilemmas faced by architects and designers.

Khatami et al. (2023, pp. 263-264) stress the rule of law and ethical behaviour within organisations, including mechanisms for promoting and monitoring ethical codes, observing ethical principles in research, and addressing conflicts of interest. Similarly, You and Lee (2011, p. 598) emphasise understanding design-related legal regulations,

including intellectual property rights, patents, copyrights, and trademarks. Furthermore, Sonneveld (2016) highlights professional ethics by developing ethical codes for designers and involving students in ethical debates.

According to Boehnert et al. (2022, pp. 8-9), critical pedagogies and decolonization deconstruct the social reproduction of dehumanising effects. Eco-pedagogy connects critical pedagogies with environmental education and social movements for inclusive social change, emphasising inclusive design. Boehnert et al. (2022, p. 9) also argue for integrating various disciplines and systemic approaches to solving complex eco-social challenges.

De Vere et al. (2009, p. 4) associate socially responsible design projects with real-world scenarios, collaboration with community organizations, and human-centred research. Similarly, Wakabayashi and Hasumi (2006, pp. 1291-1292), associate socially responsible design with situated learning principles, including proximity to design sites, communication, community participation, and multisensory learning to enhance students' understanding and ethical decision-making skills.

The "what" dimension suggests integrating ecological literacy, empathy, and interdisciplinary collaboration into design education curricula that are covered by traditional sustainable design, UX, or HCD approaches. It emphasises design practice's potential to address sustainability challenges and inclusive social change.

For which results?

When we claim to educate responsibility in design, then what does it entail, what type of results are we hoping to see and how might they be assessed?

Participants in the study questioned the nature of responsibility and ethics in design education, seeking clarity on the components and implications of ethical practice. One participant queried, "A responsible design education program... what does it entail? What are the components? How do these components or aspects collide or interchange or relate to each other? And then what are your role and your responsibility and agency considering these?" (Transcript 3, 56:54 to 57:17). This query emphasises clearly understanding responsible design education's key elements and outcomes.

Another participant questioned whether responsible design education should take the form of prescriptive instruction or individualized approaches to ethical development. They remarked, "Is it an instructional form? So ‘here is what you should do in order to become responsible,’ or is it something that you develop, and you spur to develop your own personal responsibility" (Transcript 1, 01:03:26 to 01:04:04). This reflection highlights the tension between standardized approaches to ethical education, as found in manifestos and codices, and the need for personalized, self-directed learning experiences equipping students to develop a readiness for adopting to changes as they emerge during a future career.

Although assessment is important in evaluating responsible design education, the scoping review did not identify any significant discussion of assessment procedures in the literature. This gap suggests a need for further research. There are several paths to follow, ranging from research-based development of a codex to assessment frameworks for responsible design or arguing for philosophical preparedness. Recent experiences in the change of responsibility attitude, on the background of the Russia-Ukraine war, suggest that formalized ideas on what responsibility is and should deliver appears to risk being quickly outdated.

How?

The "How?" dimension considers the methodologies and pedagogical approaches employed in responsible design education.

Participants highlighted the diverse range of teaching methods used in responsible design education. One participant noted, "That's the whole systemic approach to understanding the problem. Why is it configured as it is right here? In terms of the history, in terms of ecology... So, it's both a design thinking approach and a classic ethnographic, but more future-oriented" (Transcript 1, 32:31 to 33:13). This reflects an interdisciplinary approach to teaching that encourages students to explore innovative solutions to complex problems.

Another participant questioned whether teachers should separate teaching approaches of responsible design education and traditional teaching approaches. They remarked, "And maybe there are new teaching approaches. Isn't that what you're objecting to? That there are some of them that are very traditional teaching approaches?... it's hard to

see it as a criterion for responsibility" (Transcript 2, 14:15 to 14:40). This suggests a willingness to reconsider conventional teaching methods in favour of more innovative and participatory approaches that support critical thinking and reflection.

Studio projects in responsible design education, according to Jonas et al. (2008, p. 1), integrate real-world applications and design research, which, as Nunes (2017, p. S1018) relates, engage students in collaborative design activities with various stakeholders (also, see Padley, 2021, p. 348). As described by Ramirez (2011, pp. 4-7), these problem-solving and innovation projects encompass a spectrum of social, environmental, and technical considerations, from creating alternative solutions to societal and environmental issues to addressing specific challenges such as crime prevention and disaster management.

Workshops and activities engage students in ethical dilemmas, according to Af Ornäs and Keitsch (2016, pp. 4-5), to support critical thinking by prompting students to identify prejudices and ethical challenges. Luján Escalante et al. (2022, pp. 8-9) and Eggink et al. (2022, pp. 3-4) argue that scenario-based discussions and hands-on activities like designing packaging and envisioning future solutions enable students to explore the complexities of responsible design, emphasising empathy and anticipation (also, see Eggink, 2021, pp. 2-3).

In conclusion, how socially responsible design is taught in practice seems to align significantly with traditional design education pedagogies, through a balancing of active social engagement with reflective, ethical considerations. As such, the meeting of socially responsible learning objectives seems to not be carried out utilising distinct and specialised design pedagogies, but rather through typical interactive design practice as well as thoughtful ethical considerations as is often carried out in design studio environments.

The "How?" dimension in responsible design education may suggest a need for innovative and participatory teaching approach frameworks that ensure the full support of critical thinking, reflection, and engagement with real-world applications to address complex societal and environmental challenges through ethical design practices. It is advisable to further explore how the pedagogical integration of ethical considerations with real-world design cases may unfold in a scaffolded manner across semesters, to

build up enhanced student capabilities towards eventually managing complex and ethically conflicted design projects. Appropriate scaffolding may require new pedagogical frameworks that consider the order in which pedagogies involving direct social engagement may meaningfully interrelate with teaching approaches aiming towards awareness and action on distal cultural and environmental issues.

Discussion

Ethics, responsibility, and dialogue link systems thinking and design thinking together, highlighting the importance of integrating these elements into responsible design education. We synthesized data from a scoping review and a participatory workshop using Fayolle and Gailly's (2008, p. 572) teaching model framework. The synthesis suggests three main tensions in responsible design education between: 1) commercial motives and ethical design praxis, 2) ethical reflection and technical production, and 3) systems thinking and design thinking.

The findings also suggest the need for learning outcomes that cultivate ethical awareness and an understanding of the broader systemic impacts of design decisions. Similarly, workshop participants questioned whether traditional professional design skills apply to responsible design activities.

The literature suggests that responsible design should educate designers with an ethical "habitus" and moral character. The participants suggested tensions between prioritizing user needs and personal ethics versus broader systemic impacts and professional ethics. Workshop participants also questioned whether ethical practice in design education should be a form of prescriptive instruction, or should support individualized approaches, or some mixture of both approaches.

The content of responsible design education may include how different ethical perspectives can inform design practice and education. Additional subject knowledge may include ecological literacy, sustainability, stakeholder values, and professional ethics codes. This content is generally taught through studio projects with real-world applications and stakeholder collaboration. Alternative pedagogy includes workshops and activities to support critical thinking through grappling with ethical dilemmas and

scenarios. Participants questioned whether traditional approaches are adequate for the task or whether new teaching methods are needed.



Figure 5. Anti-Hero Card Deck created by Sai Shruthi Chivukula, Shikha Mehta, and Colin M. Grey and licensed under CC BY-NC-SA 4.0

The Anti-Hero Card Deck (Figure 5) is an example of a method that integrates ethics, responsibility, and dialogue into design (Mehta et al., 2024). The cards prompt ethical dialogue that exposes the coercive roles that designers can take during the design process. The method uses contrasting "anti-hero" and "hero" roles to nudge designers to reflect on the ethical implications of their design decisions and to avoid unethical practices that prioritize commercial interests over user values. This approach resonates with Brookfield and Preskill's (2005, p. 89-91) recommendation to use conversation roles in discussion groups to support students in learning to do communicative action (also, see Feast, 2020, p. 1889).

Although the scoping review aimed to capture a comprehensive understanding of responsible design education, the search parameters may have excluded relevant literature from certain disciplines or geographic regions. For example, 50% of studies

included in the review had first authors affiliated with institutions in the USA, Australia, or the UK. This suggests an Anglo-Saxon and Global North bias. Furthermore, the search required the keyword “design education” in the full text, which tends to exclude engineering ethics from the sample. Future studies could expand the search criteria to ensure a more inclusive representation of perspectives from more disciplines or geographic regions.

The participatory workshop involved design academics from educational institutions within a single country. Expanding the participant pool to include a more diverse range of stakeholders from various countries and cultural backgrounds could provide additional insights and perspectives.

The findings indicate a potential gap in the literature regarding assessment procedures and frameworks for responsible design education. A further systematic literature review, specifically targeting the assessment of responsible design education is needed to establish the scope of this gap in the literature.

Future research could also investigate the connections between various factors, such as design sub-domains, degree types, design paradigms, institutional contexts, and geographic regions, in shaping social accountability in design education and professional practice.

Conclusion

We began this article by observing the importance of integrating ethics, responsibility, and dialogue into design education to address complex designerly and socio-ecological challenges. We found that responsible design education should cultivate both ethical awareness and a broad understanding of the systemic impacts of design decisions. The value of this perspective is that it promotes the development of an ethical character among design students, equipping them to navigate complex ethical dilemmas and prioritize responsibility over commercial motives.

The issues discussed give us better insight into the nature of responsible design as an approach that emphasises praxis, ethical reflection, and systems thinking rather than solely technical production. Additionally, responsible design education should encompass interdisciplinary collaboration, ecological literacy, stakeholder value

consideration, ethical perspectives, professional ethics codes and personal values. Furthermore, the discussion highlights the role of studio projects with real-world applications as well as the significance of innovative pedagogies, such as workshops supporting critical thinking through ethical dilemmas.

This paper's contribution has been to highlight the scope, content, and pedagogical approaches required to integrate ethics, responsibility, and dialogue into design education. While the paper points to some immediate research avenues to be pursued, there are no silver bullets presented to ensure responsibility in future design and design students' praxis beyond being philosophically prepared for difficult and ever-changing agendas in what may be responsible acts. Additionally, caveats worth mentioning include the potential Anglo-Saxon and Global North bias in the literature reviewed and the need for broader stakeholder perspectives. We do, however, hope that by making these arguments, we have made clearer the scope and meaning of responsible design education.

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References

1. Af Ornäs, V. H., & Keitsch, M. (2016). Ethics in Design Curricula: Teaching Approaches. *18th International Conference on Engineering and Product Design Education (E&PDE)*, Aalborg University, Aalborg, Denmark.
2. Boehm, Sarah. (2015). Facilitating Comprehension, Connection and Commitment to Environmentally Responsible Design. *International Journal of Art & Design Education*, *34*(1), 73-88. https://doi.org/10.1111/jade.12010
3. Boehnert, Joanna, Sinclair, Matt, & Dewberry, Emma. (2022). Sustainable and Responsible Design Education: Tensions in Transitions. *Sustainability*, *14*(11). https://doi.org/10.3390/su14116397
4. Bousbaci, Rabah, & Findeli, Alain. (2005). More Acting and Less Making: A Place for Ethics in Architecture's Epistemology. Design *Philosophy Papers*, *3*(4), 245-264. https://doi.org/10.2752/144871305X13966254124996

5. Brookfield, Stephen D., & Preskill, Stephen. (2005). *Discussion as a way of teaching: Tools and techniques for democratic classrooms* (Second edition. ed.). Jossey-Bass.
6. Chan, Jeffrey. (2016). From Afterthought to Precondition: Re-engaging Design Ethics from Technology, Sustainability, and Responsibility. *Future Focused Thinking - DRS International Conference 2016*, Brighton, United Kingdom.
7. Christensen, B. T., & Ball, L. J. (2019). Building a discipline: Indicators of expansion, integration and consolidation in design research across four decades. *Design Studies*, *65*, 18-34.
8. Cipolla, C., & Bartholo, R. (2014). Empathy or Inclusion: A Dialogical Approach to Socially Responsible Design. *International Journal of Design*, *8*(2), 87-100.
9. Cross, Nigel. (1972). *Design Participation - Proceedings of the Design Research Society Conference, 1971*. Design Research Society. https://dl.designresearchsociety.org/conference-volumes/1
10. Culén, Alma Leora, & Karahasanovic, Amela. (2022). Towards responsible interaction design education. *DRS2022: Bilbao*, Bilbao, Spain.
11. d'Anjou, Philippe. (2004). Theoretical and methodological elements for integrating ethics as a foundation into the education of professional and design disciplines. *Science and Engineering Ethics, 10*(2), 211-218. https://doi.org/10.1007/s11948-004-0016-6
12. Das, M., Roeder, G., Ostrowski, A. K., Yang, M. C., & Verma, A. (2023). What Do We Mean When We Write About Ethics, Equity, and Justice in Engineering Design? *Journal of Mechanical Design, 145*(6). https://doi.org/10.1115/1.4057056
13. Davis, Meredith, & Dubberly, Hugh. (2023). Rethinking Design Education. *She Ji: The Journal of Design, Economics, and Innovation, 9*(2), 97-116. https://doi.org/https://doi.org/10.1016/j.sheji.2023.04.003
14. de Vere, Ian, Bissett-Johnson, Katherine, & Thong, Christine. (2009). Educating the responsible engineer; socially responsible design and sustainability in the curriculum. *DS 59: Proceedings of E&PDE 2009, the 11th Engineering and Product Design Education Conference-Creating a Better World*, Brighton, UK, 10.-11.09. 2009,
15. Debs, L., Gray, C. M., & Asunda, P. A. (2023). Students' perceptions and reasoning patterns about the ethics of emerging technology. *International Journal of Technology and Design Education, 33*(1), 143-163. https://doi.org/10.1007/s10798-021-09719-w

16. Donahue, Sean. (2004). Discipline Specific Ethics. *Design Philosophy Papers, 2*(2), 95-101. https://doi.org/10.2752/144871304X13966215067877
17. Eggink, W., Ozkaramanli, D., Zaga, C., & Liberati, N. (2020). Setting the stage for Responsible Design. In *Design Research Society, DRS 2020: Synergy*. Design Research Society.
18. Eggink, Wouter. (2021). Ethics in Design Education, but different: Practicing the Practical Turn. *DS 110: Proceedings of the 23rd International Conference on Engineering and Product Design Education (E&PDE 2021)*, VIA Design, VIA University in Herning, Denmark. 9th-10th September 2021,
19. Eggink, Wouter, Dorrestijn, Steven, van der Heijden, Karin, & Ouwens, Ilse. (2022). Ethics, design, and creativity: A fruitful combination. *DRS2022: Bilbao*, Bilbao, Spain.
20. Escobar, A. (2018). *Designs for the Pluriverse*. Duke University Press.
21. Elliott, Jack. (2004). Reconciling Eco-Ethics and Aesthetics. *Design Philosophy Papers, 2*(2), 115-128. https://doi.org/10.2752/144871304X13966215067958
22. Fayolle, Alain, & Gailly, Benoit. (2008). From craft to science. *Journal of European Industrial Training, 32*(7), 569-593. https://doi.org/10.1108/03090590810899838
23. Feast, Luke. (2020). Design Education for the Knowledge Society: An Action Research Study of Implementing a Liberal Arts Approach to Industrial Design Education. *DRS2020: Syner*gy, Brisbane.
24. Ferrarello, Laura. (2019). Social awareness in design & engineering education and practice: the value of ethics in postgraduate education. *International conference on engineering and product design education*, United Kingdom.
25. Findeli, A. (2001). Rethinking design education for the 21st century: Theoretical, methodological, and ethical discussion. *Design Issues, 17*(1), 5-17. https://doi.org/10.1162/07479360152103796
26. Fry, Tony. (2004). The Voice of Sustainment: Design Ethics as Futuring. *Design Philosophy Papers, 2*(2), 145-156. https://doi.org/10.2752/144871304X13966215068038
27. Gemser, Gerda, de Bont, Cees, Hekkert, Paul, & Friedman, Ken. (2012). Quality perceptions of design journals: The design scholars' perspective. *Design Studies, 33*(1), 4-23. https://doi.org/10.1016/j.destud.2011.09.001
28. Godbold, Rosemary, Lees, Amanda, & Reay, Stephen. (2019). Ethical Challenges for Student Design Projects in Health Care Settings in New Zealand. *International Journal*

of Art & Design Education, 38(1), 182-192. https://doi.org/https://doi.org/10.1111/jade.12170

29. Gray, C. M. (2019). Revealing Students' Ethical Awareness during Problem Framing. *International Journal of Art & Design Education, 38*(2), 299-313. https://doi.org/10.1111/jade.12190
30. Gray, C. M., & Boling, E. (2016). Inscribing ethics and values in designs for learning: a problematic. *ETR&D Educational Technology Research and Development, 64*(5), 969-1001. https://doi.org/10.1007/s11423-016-9478-x
31. Gray, Colin M., Chivukula, Shruthi Sai, Carlock, Thomas V., Li, Ziqing, & Duane, Ja-Nae. (2023). Scaffolding Ethics-Focused Methods for Practice Resonance. *Proceedings of the 2023 ACM Designing Interactive Systems Conferenc*e, Pittsburgh.
32. Haug, A. (2017). Educating ethical designers. *International Journal of Technology and Design Education, 27*(4), 655-665. https://doi.org/10.1007/s10798-016-9367-y
33. Jonas, Wolfgang, Morelli, Nicola, & Munch, Juliane. (2008). Designing a product service system in a social framework: methodological and ethical considerations. *Undisciplined! - DRS International Conference 2008*, Sheffield, United Kingdom.
34. Khaniki, Z., & Khalili, M. (2017). The Role of Behavior Framing Factors in Industrial Design Education to Inspire the Responsible Attitude. *19th International Conference on Engineering and Product Design Education (E&PDE)*, Akershus Univ Coll Appl Sci, Oslo, NORWAY.
35. Khatami, S. M., Boujari, P., & Ranjbar, E. (2023). Toward a social responsibility-based model for urban design education. *Urban Design International, 28*(4), 256-271. https://doi.org/10.1057/s41289-022-00195-9
36. Krogh, P., & Koskinen, I. (2020). *Drifting by intention*. Dordrecht, the Netherlands: Springer.
37. Leerberg, M., Riisberg, V., & Boutrup, J. (2010). Design Responsibility and Sustainable Design as Reflective Practice: An Educational Challenge. *Sustainable Development, 18*(5), 306-317. https://doi.org/10.1002/sd.481
38. Lloyd, P., & van de Poel, I. (2008). Designing games to teach ethics. *Science and Engineering Ethics, 14*(3), 433-447. https://doi.org/10.1007/s11948-008-9077-2
39. Luján Escalante, Maria Alejandra, Moffat, Luke, & Büscher, Monika. (2022). Ethics through design. *DRS2022: Bilbao*, Bilbao, Spain.

40. Mehta, Shikha, Chivukula, Shruthi S., Gray, Colin M., & Gairola, Ritika. (2024). Anti-Heroes: An ethics-focused method for responsible designer intentions. arXiv [cs.HC]. http://arxiv.org/abs/2405.03674
41. Melles, G., de Vere, I., & Misic, V. (2011). Socially responsible design: thinking beyond the triple bottom line to socially responsive and sustainable product design. *Codesign-International Journal of Cocreation in Design and the Arts, 7*(3-4), 143-154. https://doi.org/10.1080/15710882.2011.630473
42. Mohamed, T. I. (2021). Refine the Students' Skills Through Design Education to Create Ethical Products. *Sage Open, 11*(1), https://doi.org/10.1177/2158244021998360
43. Moreno, M., De los Rios, C., Rowe, Z., & Charnley, F. (2016). A conceptual framework for circular design. *Sustainability, 8*(9), 937.
44. Nunes, V. G. A. (2017). Designing more responsible behaviours through Design Education: Reflections on a Brazilian pilot experience in Social Innovation for Sustainability. *Design Journal, 20*, S1014-S1025. https://doi.org/10.1080/14606925.2017.1353045
45. Ozkaramanli, Deger, Nagenborg, Michael, Fantini van Ditmar, Delfina, Lehtinen, Sanna, Schwobel-Patel, Christine, & Ferrarello, Laura. (2022). Design + Ethics: How is it more than the sum of its parts? *DRS2022: Bilbao*, Bilbao, Spain.
46. Ozkaramanli, D., Ferrarello, L.F., and Laursen, L.N. (2024) Ethics in/of/for Design, in Gray, C., Ciliotta Chehade, E., Hekkert, P., Forlano, L., Ciuccarelli, P., Lloyd, P. (eds.), *DRS2024: Boston*, 23–28 June, Boston, USA. https://doi.org/10.21606/drs.2024.159
47. Padley, A. (2021). The agile ethics process: An oxymoron or a new paradigm in education? *Industry and Higher Education,* 35(4), 347-352. https://doi.org/10.1177/09504222211015606
48. Papanek, Victor. (2000). *Design for the real world: Human ecology and social change* (2nd ed.). Academy Chicago. (1971)
49. Pask, G., Jones, J. C., & Thornley, D. G. (1962). The conception of a shape and the evolution of a design. *The conference on design methods* (p. 153).
50. Ramirez, Mariano. (2011). Designing with a social conscience: An emerging area in industrial design education and practice. *DS 68-5: Proceedings of the 18th International Conference on Engineering Design (ICED 11), Impacting Society through*

Engineering Design, Vol. 5: Design for X/Design to X, Lyngby/Copenhagen, Denmark, 15.-19.08. 2011,

51. Schon, D. A. (1968). *The reflective practitioner*. New York.
52. Simon, H. A. (1981). *The sciences of the artificial*, 1969. Massachusetts Institute of Technology.
53. Sonneveld, M. H. (2014). Positive Ethics in Design Education. *16th International Conference on Engineering and Product Design Education (E&PDE)*, Univ Twente, Enschede, NETHERLANDS.
54. Sonneveld, M. H. (2016). Ethics in Design Education: An Integrated Approach. *18th International Conference on Engineering and Product Design Education (E&PDE)*, Aalborg Univ, Aalborg, DENMARK.
55. Stewart, Susan, & Lorber-Kasunic, Jacqueline. (2006). Akrasia, Ethics and Design Education. *Design Philosophy Papers, 4*(4), 231-245. https://doi.org/10.2752/144871306X13966268131910
56. Tollestrup, C., Laursen, L. N., Vesti, H. N. (2023) 'Design Expertise: A Structured Literature Review', in *Proceedings of the International Conference on Engineering Design (ICED23)*, Bordeaux, France, 24-28 July 2023. DOI:10.1017/pds.2023.124
57. Tricco, Andrea C., Lillie, Erin, & Zarin et al, Wasifa. (2018). PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation. *Annals of Internal Medicine, 169(*7), 467-473.
58. Vesti, H., N. Laursen, L., and Tollestrup, C. (2024) Past, present and future of design ethics, in Gray, C., Ciliotta Chehade, E., Hekkert, P., Forlano, L., Ciuccarelli, P., Lloyd, P. (eds.), *DRS2024: Boston*, 23–28 June, Boston, USA. https://doi.org/10.21606/drs.2024.844
59. von Foerster, Heinz. (2003). *Understanding Understanding: Essays on Cybernetics and Cognition*. Springer.
60. Wakabayashi, N., & Hasumi, T. (2006). Application of Situated Learning for Realizing Design Ethics. *Design Society 9th International Design Conference,* Dubrovnik, CROATIA.
61. Wakkary, R. (2021). *Things we could design: For more than human-centred worlds*. MIT Press.

62. Watkins, Christopher Rhys, Gray, Colin M., Toombs, Austin L., & Parsons, Paul. (2020). Tensions in Enacting a Design Philosophy in UX Practice. *Proceedings of the 2020 ACM Designing Interactive Systems Conference*
63. You, M. L., & Lee, Y. Y. (2011). Design Ethics Education in Taiwan: A Study of Syllabi of Ethics-Related Courses. *4th International Conference on Internationalization, Design and Global Development*, Orlando, FL.
64. Yürekli, I. (2015). Ethics and Conscience: An Integral Part of Architectural Design Education. *Open House International, 40*(3), 76-83.

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Appendix

Search Parameters

1. Title= Responsib* or ethic*
2. Topic= “Design education”
3. Publication date = 1999 to 2024
4. Language = English
5. Document type = Article OR Proceedings Papers
6. #1 AND #2 AND #3 AND #4 AND #5

Targeted journals:

Applied Ergonomics

Art Design and Communication in Higher Education

CAD Computer-Aided Design

CoDesign

Creativity and Innovation Management

Design And Culture

Design Issues

Design Management Journal

Design Philosophy Papers

Design Research Quarterly

Design Science

Design Studies

Digital Creativity

Empirical Studies of the Arts

Ergonomics

Ergonomics in design

Human Factors

Human-Computer Interaction

Information Design Journal

Interacting with Computers

International Journal of Art and Design Education

International Journal of Design

International Journal of Product Development

International Journal of Sustainable Design

International Journal of Technology and Design Education

International Journal on Interactive Design and Manufacturing

Journal of Design History

Journal of Design Research

Journal of Engineering Design

Journal of Engineering Education

Journal of Interior Design

Journal of Material Culture

Journal of Mechanical Design [transactions of the ASME]

Journal of Product Innovation Management

Journal on Multimodal User Interfaces

Materials and Design

Personal and Ubiquitous Computing

Research in Engineering Design

She Ji

The Design Journal

The Journal of Sustainable Product Design

The Senses & Society

Visible Language

Visual Communication

Targeted conferences:

Design Research Society

Nordic Design Research Conference

ACM Conference on Designing Interactive Systems

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